

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 04 – Discriminative Classifiers I: Linear, Generalized, Piecewise, and Fisher



Chizhi Chris ZHANG

zhangchizhi@ciomp.ac.cn

Advanced Computing and Digital Technology Research Center

University of Chinese Academy of Sciences

reaches the sorting gate

The observation

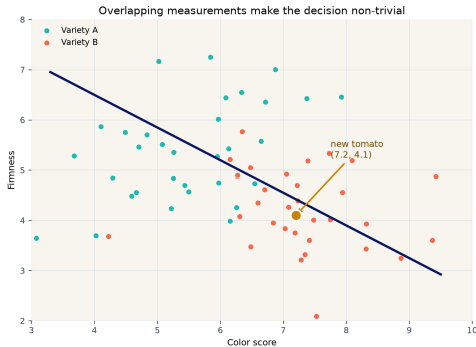
Color score $x_1 = 7.2$ and firmness $x_2 = 4.1$.

The decision

Pack now or hold for another day?

The complication

Two varieties overlap in the sensor space.



The same data support two questions

Question	Quantity	Typical use
How could this measurement arise?	$p(x \mid y)$ and $p(y)$	Describe each class
Which label fits this measurement?	$p(y \mid x)$	Predict a class
Which side of the boundary is it on?	$g(x)$	Make a decision

One data table. Different questions lead to different models.

Lecture map

先进计算与数字工程研究中心

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Syllabus topic	Question answered
Classifier families	What should the model represent?
Linear discriminant functions	What does one straight boundary mean?
Generalized linear functions	How can a feature map create a curve?
Piecewise linear functions	How can several local lines form one rule?
Fisher linear discriminant	Which projection separates the classes?



Programming

NumPy examples will calculate scores, maps, prototypes, and Fisher directions.

Model and decision model

Generative route

Fit $p(x | y)$ and $p(y)$, then derive $p(y | x)$.

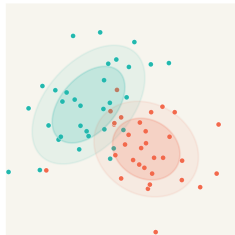
Discriminative route

Fit $p(y | x)$ or a score $g(x)$ directly.

Do not rank them by name

The useful choice depends on data, assumptions, and the required output.

Generative: describe each class	Discriminative: describe the decision
--	--

 $p(x | y)$ and $p(y)$

Discriminative: describe the decision

 $p(y | x)$ or $q(x)$

Generative modelling describes each class

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Tomato example

Fit a color and firmness distribution for each maturity class.

$$p(y = k | x) = \frac{p(x | y = k)p(y = k)}{\sum_j p(x | y = j)p(y = j)}$$



What the model can do. Classify, sample plausible measurements, and reveal unusual observations.

Discriminative modelling targets the label

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

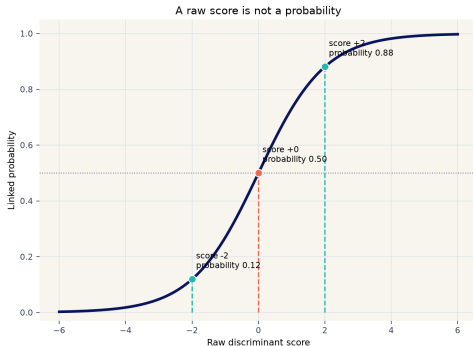
Conditional probability

Estimate $p(y | x)$ when calibrated uncertainty matters.

Decision score

Estimate $g(x)$ when only the class decision is required.

$$\hat{y} = \arg \max_k p(y = k | x) \quad \text{or} \quad \hat{y} = \text{sign } g(x)$$



Choose class 1 when $p(y = 1 | x) > p(y = 2 | x)$.

$$g(x) = \log p(x \mid y = 1) + \log p(y = 1) - \log p(x \mid y = 2) - \log p(y = 2)$$

Class 1 is selected when $g(x) > 0$.

A wrong mixture or cluster assumption can move the boundary in the wrong direction.

Without an added objective, unlabeled rows contribute no label loss.

model
ing entries
servations

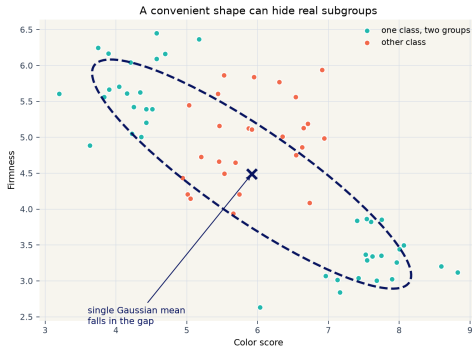
A Gaussian ellipse cannot represent every class distribution.

Hidden groups

Two tomato varieties may create two modes inside the same maturity class.

Consequence

The estimated likelihood can be wrong far from the boundary and still alter the decision.



Comparing one tuned model with another model's default settings proves little.

family choice 工程研究中心



Thirty second answer

Choose one row and state the assumption that makes your choice reasonable.

The distinction to keep

先进计算与数字工程研究中心

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Generative

Model how x relates to each class, then derive the decision.

Discriminative

Model the conditional label or decision function directly.

Evaluation

The test set decides whether the chosen assumptions helped.

Discriminant functions

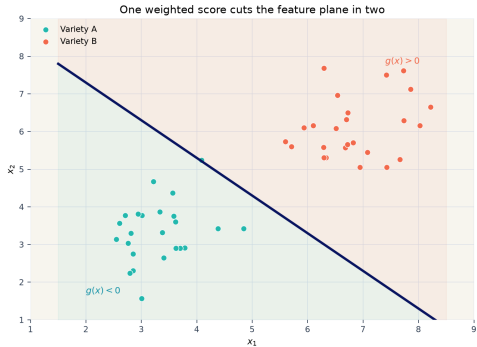
The model

A weighted sum turns a feature vector into one score.

$$g(x) = w^\top x + w_0$$

The boundary

All points satisfying $g(x) = 0$.



A score turns measurements into a decision

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Two class rule

Predict class +1 when $g(x) > 0$. Predict class -1 when $g(x) < 0$.

At exactly zero

Use a fixed tie rule or send the sample for review.

A score is not a label

The sign creates the label. The magnitude describes confidence only under extra assumptions.

Two features make a line

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

$$g(x) = w_1 x_1 + w_2 x_2 + w_0$$

Decision boundary

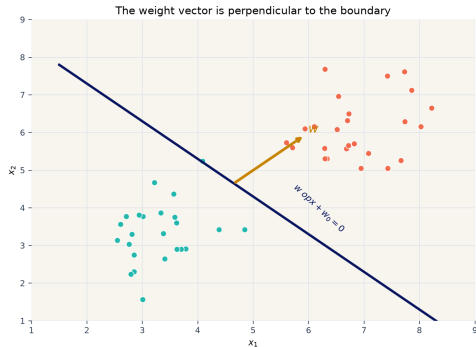
$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

If $w_2 \neq 0$

$$x_2 = -\frac{w_1}{w_2} x_1 - \frac{w_0}{w_2}$$

Geometric meaning. Slope $-w_1/w_2$ and vertical intercept $-w_0/w_2$.

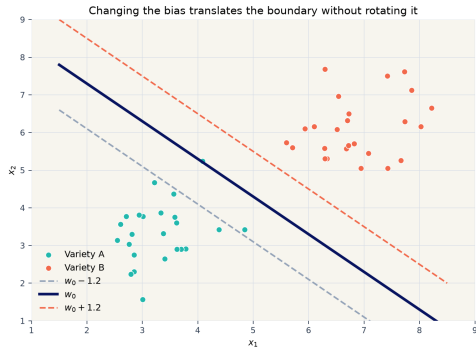
Moving in the direction w increases $g(x)$ fastest.



Changing w_0 moves the boundary parallel to itself.

$$\text{shift along } w = -\frac{\Delta w_0}{\|w\|}$$

A constant color offset can often be absorbed by the bias.



Signed score and geometric distance

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Raw score

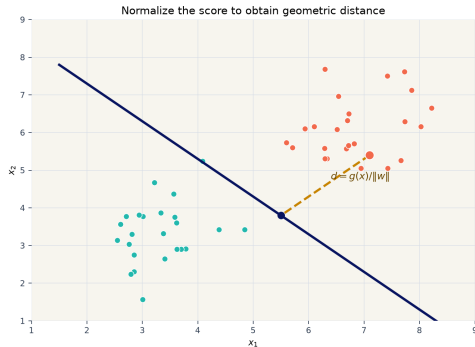
$g(x) = w^T x + w_0$ changes if every parameter is rescaled.

Signed distance

$$d(x) = \frac{w^T x + w_0}{\|w\|}$$

Why divide

$g(x)$ and $10g(x)$ define the same boundary but not the same raw magnitude.



ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

$$w = (1, -0.8)^\top, w_0 = -3.5, x = (7.2, 4.1)^\top$$

$$g(x) = 7.2 - 0.8 \times 4.1 - 3.5 = 0.42$$

The score is positive, so the model selects pack now.

Distance. $0.42/\sqrt{1^2 + 0.8^2} \approx 0.328$ feature units.

Standardization makes scales comparable

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Transform each feature

$$z_j = \frac{x_j - \mu_j}{s_j}$$

Interpretation

A one unit change in z_j means one training standard deviation.

Deployment rule

Reuse the training values of μ_j and s_j for new samples.

The augmented vector absorbs the bias

Augmented feature

$$\tilde{X} = (X_1, \dots, X_D, 1)^\top$$

Augmented weight

$$\tilde{W} = (w_1, \dots, w_D, w_0)^\top$$

$$g(x) = \tilde{w}^\top \tilde{x}$$

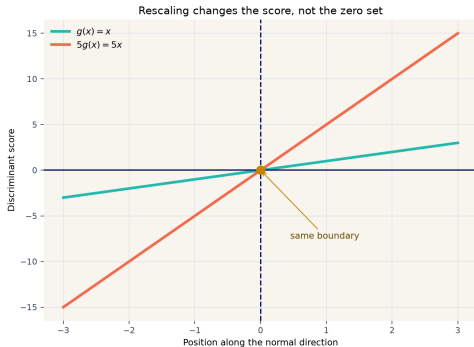
Why it helps

One inner product handles both feature weights and the bias.

(w, w_0) and (cw, cw_0) define the same decision for every $c > 0$.

For $c < 0$, the boundary stays but the class labels swap.

A normalization or training objective selects one representative from many equivalent parameter sets.



Separable training points do not guarantee perfect predictions on future data.

Exclusive or cannot use one straight boundary

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

x_1	x_2	Positive class
0	0	Yes
0	1	No
1	0	No
1	1	Yes



Contradiction

No affine function can place both diagonal corners on one side and the other corners on the opposite side.

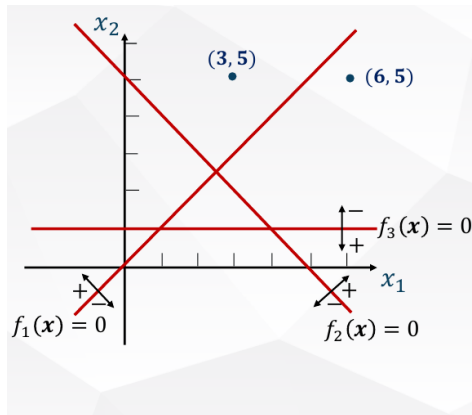
Later solutions

Add an interaction feature, combine local lines, or use a deeper model.

$g_c(x) > 0$ means class c versus all other classes.

$$g_c(x) = w_c^\top x + w_{c0}, \quad c = 1, \dots, C$$

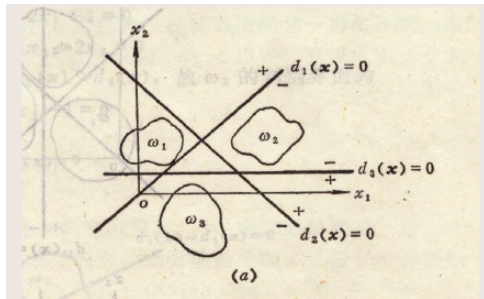
Fit separate tests for premium, standard, and hold.



Several classes claim the same sample.

No classifier claims the sample.

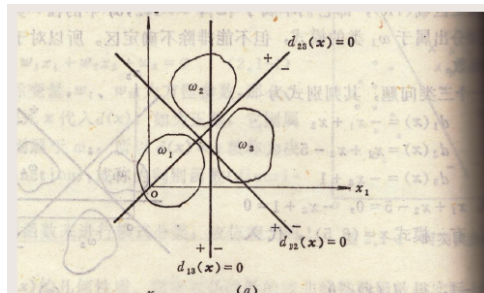
Use comparable scores, a priority rule, or a reject option. State the choice explicitly.



Fit one classifier for every pair of classes.

$$\text{number of classifiers} = \frac{C(C-1)}{2}$$

Each pair casts one vote. The class with most votes wins.



Pairwise voting can tie

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Three pairwise results

A beats B, B beats C, and C beats A.

Class	Votes
A	1
B	1
C	1



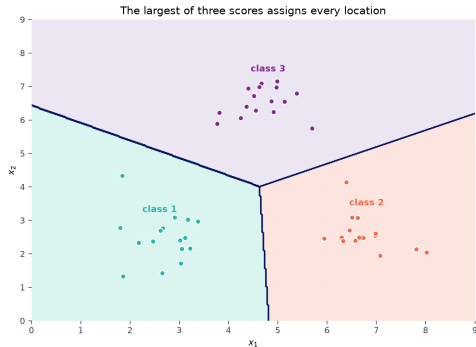
A complete classifier needs a tie rule

Examples include summed margins, calibrated pairwise probabilities, or a fixed class order.

Fit C scores and choose the largest.

$$\hat{y} = \arg \max_c g_c(x), \quad g_c(x) = w_c^\top x + w_{c0}$$

Classes c and k meet where $g_c(x) = g_k(x)$.



Argmax partitions the whole plane

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

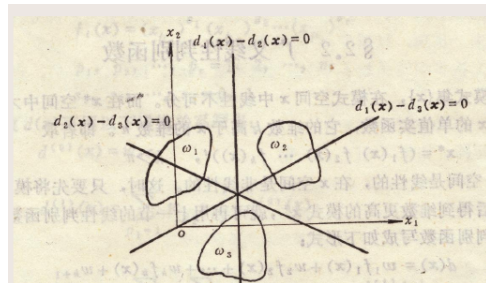
Complete assignment

Every point has at least one largest score.

Ties

Two or more classes can share the maximum exactly on a boundary.

Geometry. The visible class regions are intersections of linear inequalities.



A numeric three class decision

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Scores

$$g_1 = -x_1 + x_2, g_2 = x_1 + x_2 - 5, g_3 = -x_2 + 1$$

Input	g_1	g_2	g_3	Argmax
(6, 5)	-1	6	-4	Class 2
(3, 5)	2	3	-4	Class 2



One versus rest difference

At (3, 5), both g_1 and g_2 are positive.

Classifier count grows differently

Classes	One versus rest	One versus one	Comparable class scores
3	3	3	3
10	10	45	10
100	100	4,950	100



Cost beyond training

Prediction must also evaluate the required scores or pairwise classifiers.

Scores and probabilities answer different questions

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Output	What it supports	What it does not guarantee
Score $g(x)$	Ranking and boundary decisions	Calibrated probability
Margin distance	Geometric separation	Frequency among similar cases
Probability $p(y x)$	Expected risk and thresholds	Correctness without validation

A link function can convert a score

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Logistic example

$$p(y = 1 \mid x) = \frac{1}{1 + \exp[-g(x)]}$$

Boundary

$g(x) = 0$ corresponds to probability 0.5.

Preview only

The link defines a probability model. Its fitting and evaluation appear in the linear models lecture.

Least squares can fit a discriminant

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Targets

Encode classes as $t_i \in \{-1, +1\}$.

Objective

$$\min_{\tilde{w}} \|\tilde{X}\tilde{w} - t\|_2^2$$

Closed form when appropriate

Solve $\tilde{X}^\top \tilde{X} \tilde{w} = \tilde{X}^\top t$.

Least squares uses every residual

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Correct but extreme

A positive sample with target $+1$ and score $+8$ has residual 7 .

Wrong but near

A positive sample with score -0.1 has residual -1.1 .

Mismatch

The first point is correctly classified but contributes a much larger squared error.

A large residual can pull the boundary

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Before the extreme point

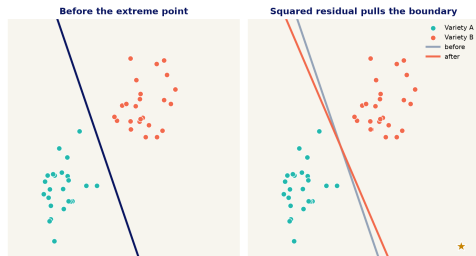
The fitted line follows the main cloud.

After the extreme point

Squared loss gives the distant residual disproportionate influence.

Response

Inspect leverage, measurement errors, and a classification specific loss.



NumPy batch scoring

数字工程研究中心

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

```
import numpy as np

X = np.array([[7.2, 4.1],
              [5.4, 6.0],
              [6.8, 3.5]])
w = np.array([1.0, -0.8])
b = -3.5

scores = X @ w + b
labels = np.where(scores > 0, 1, -1)
print(scores)
print(labels)
```

Expected scores

[0.42, -2.90, 0.50] and labels [1, -1, 1].

NumPy multiclass scores

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

```
W = np.array([[ -1.,  1.],
               [ 1.,  1.],
               [ 0., -1.]])
b = np.array([0., -5., 1.])
X = np.array([[6., 5.], [3., 5.]])

scores = X @ W.T + b
prediction = scores.argmax(axis=1) + 1
print(scores)
print(prediction)
```

Axis check

Rows are samples and columns are class scores. Take the maximum across columns.

Linear model checkpoint

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Boundary

$$g(x) = 2x_1 - x_2 - 3$$

- 1 State the normal vector and the slope.
- 2 Classify $x = (2, 0)^\top$.
- 3 Compute its signed distance to the boundary.
- 4 Predict what changes if every parameter is multiplied by -2 .



Answers

$w = (2, -1)^\top$, slope 2, score 1, distance $1/\sqrt{5}$. The boundary stays and labels swap.

Main idea

Transform x into features $\phi(x)$, then use a linear score in the transformed space.

Result

A hyperplane in feature space can appear curved in the original space.

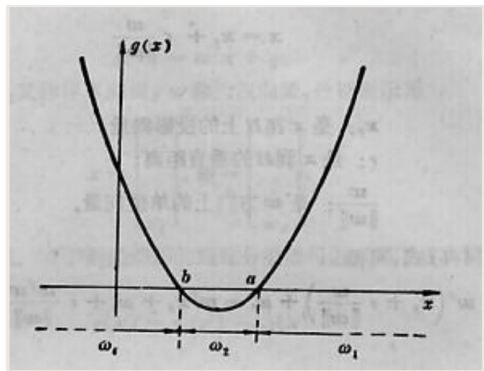
If $\phi_1(x) = x^2$, the score is nonlinear in x .

Generalized linear here means a fixed feature map followed by a linear discriminant. It is distinct from the statistical family called generalized linear models.

Class 2 occupies $b < x < a$. Class 1 occupies $x \leq b$ or $x \geq a$.

A one dimensional affine score changes sign at most once.

The desired rule needs two crossing points.



Let $z_1 = x^2$ and $z_2 = x$.

$$z_1 - (a + b)z_2 + ab = 0$$

This is a line in the z plane and a quadratic equation on the original x axis.

Exclusive or becomes linear with one interaction

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

x_1	x_2	x_1x_2	Target y
-1	-1	+1	+1
-1	+1	-1	-1
+1	-1	-1	-1
+1	+1	+1	+1



A one feature rule

$$g(x) = x_1x_2 \text{ and } \hat{y} = \text{sign } g(x).$$

What changed

The interaction records whether the two signs agree.

$$\phi(x) = (x_1^2, x_1x_2, x_2^2, x_1, x_2, 1)^\top$$

$$g(x) = w_{11}x_1^2 + w_{12}x_1x_2 + w_{22}x_2^2 + w_1x_1 + w_2x_2 + w_0$$

Ellipses, parabolas, hyperbolas, pairs of lines, and degenerate cases.

Polynomial degree grows the feature space

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

All monomials up to degree r

$$M = \binom{D+r}{r}$$

Input dimension D	Degree r	Features M
10	2	66
10	3	286
100	2	5,151
100	3	176,851



Capacity warning

Higher degree can overwhelm both memory and the available sample size.

Powers amplify scale differences

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Two raw features

Color ranges near 10 while acidity ranges near 0.1.

After squaring

The scales become about 100 and 0.01.

Numerical effect

Optimization and regularization can become dominated by feature units.

Usual response. Standardize suitable inputs before constructing polynomial terms.

The boundary may oscillate in regions with little evidence.

Use validation performance, regularization, and a degree chosen before looking at the test set.

quickly
must be chosen
on preprocessing

Polynomial features by hand

```
import numpy as np

X = np.array([[1., 2.],
              [3., 4.]])
x1, x2 = X[:, 0], X[:, 1]
Phi = np.column_stack([
    x1**2, x1*x2, x2**2, x1, x2, np.ones(len(X))
])
print(Phi)
print(Phi.shape)
```

Shape

Two samples become two rows with six quadratic features.

Interval classifier in code

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

```
a, b = 3.0, 1.0
x = np.array([0.0, 1.5, 2.0, 3.5])
Phi = np.column_stack([x**2, x, np.ones(len(x))])
w = np.array([1.0, -(a + b), a * b])

scores = Phi @ w
labels = np.where(scores >= 0, 1, 2)
print(scores)
print(labels)
```

Expected labels

Class 1, class 2, class 2, class 1.

Generalized linear checkpoint

Desired rule

Class 1 when $|x| > 2$. Class 2 when $|x| < 2$.

- 1 Choose a feature map.
- 2 Write a linear score in that feature space.
- 3 State the boundary points and the tie rule.

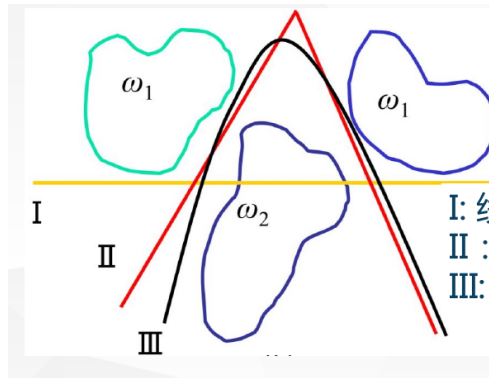
One answer

$\phi(x) = (x^2, 1)$, $g(x) = x^2 - 4$. Boundary at $x = \pm 2$.

Use several affine pieces, each responsible for a local part of the input space.

Straight segments join to approximate a curved or multi region decision surface.

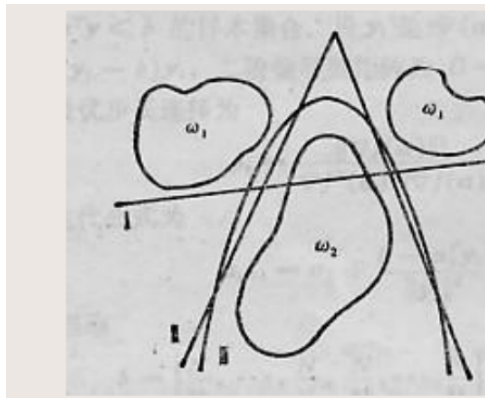
The number and placement of pieces must be learned or chosen.



The pack now class forms two separated clusters because the varieties mature at different colors.

It may cut through the hold cluster or miss one pack cluster.

Each cluster can receive a nearby linear comparison.



Boundary family	Strength	Main limitation
One affine boundary	Simple and stable	Cannot bend
Feature mapped boundary	Smooth global curvature	Feature count may grow
Piecewise affine boundary	Local flexibility	Pieces and joins need control

Control

$$g_c(x) = \max_{\ell=1, \dots, L_c} g_{c\ell}(x)$$

$$\hat{y} = \arg \max_c g_c(x)$$

pieces meet on a line

Switching set

$$g_{c1}(x) = g_{c2}(x)$$

$$(w_{c1} - w_{c2})^T x + (b_{c1} - b_{c2}) = 0$$

Meaning

The active piece changes across this hyperplane.

Not always a class boundary

Both pieces may belong to the same class.

Nearest prototypes create affine comparisons

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Choose the nearest prototype

$$\hat{j} = \arg \min_j \|x - \mu_j\|^2$$

$$-\|x - \mu_j\|^2 = -x^\top x + 2\mu_j^\top x - \|\mu_j\|^2$$



Shared term

$-x^\top x$ cancels when prototypes are compared.

Several prototypes bend the class boundary

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

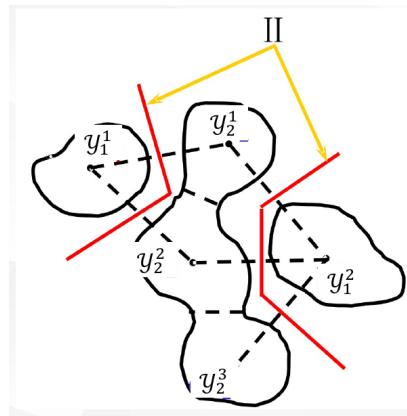
Subclasses

Class c has prototypes $\mu_{c1}, \dots, \mu_{cL_c}$.

$$d_c(x) = \min_{\ell} \|x - \mu_{c\ell}\|^2, \quad \hat{y} = \arg \min_c d_c(x)$$

Geometry

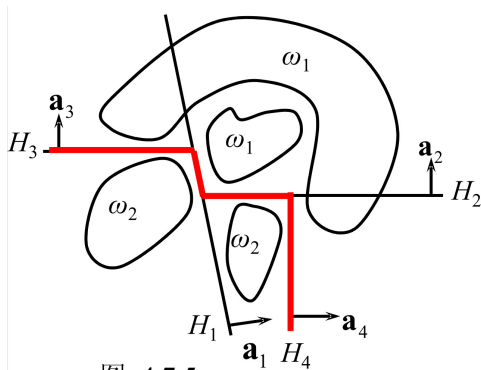
Prototype Voronoi cells meet along line segments. Class labels merge neighboring cells.



Find one difficult local comparison between opposite classes.

Add another separating plane where errors remain.

End when validation improvement no longer justifies another piece.



Choose the piece count with validation evidence, not training error alone.

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

100%

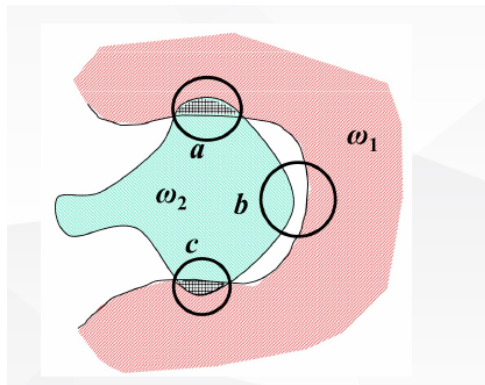
100

Merge prototypes, require minimum support, or penalize model complexity.

A local area where samples or prototypes from different classes are close.

Use nearby opposite class evidence to design the local separator.

Easy interior points still matter for defining where the local rule is allowed to act.



Prototype pairs suggest initial local planes

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

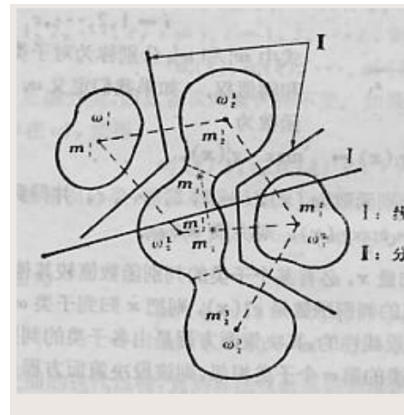
Mutual nearest prototypes

A prototype from each class is the other's nearest opposite class prototype.

$$\left[x - \frac{\mu_a + \mu_b}{2} \right]^\top (\mu_a - \mu_b) = 0$$

Initial plane

Use the perpendicular bisector, then refine with nearby samples.



The distance matrix has three samples by four prototypes.

Piecewise linear checkpoint

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Model

Class 1 has prototypes $(0, 0)$ and $(4, 0)$. Class 2 has prototype $(2, 3)$.

- 1 Write the nearest prototype rule.
- 2 Find the boundary between $(0, 0)$ and $(2, 3)$.
- 3 Explain why the full class boundary has more than one segment.



One boundary

$$x_1(2) + x_2(3) = \frac{1}{2}(2^2 + 3^2) = 6.5.$$

ear discriminant

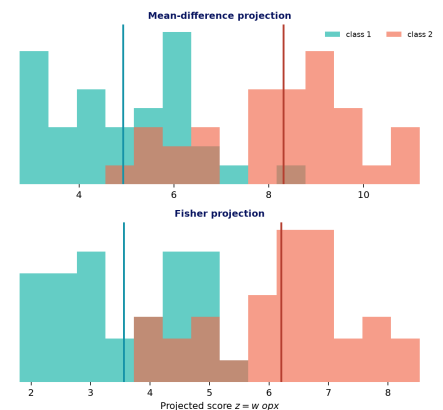
Goal

Project many features onto one axis while preserving class separation.

$$z = w^T x$$

Fisher criterion

Projected class centers should be far apart and each class should remain compact.



Same points

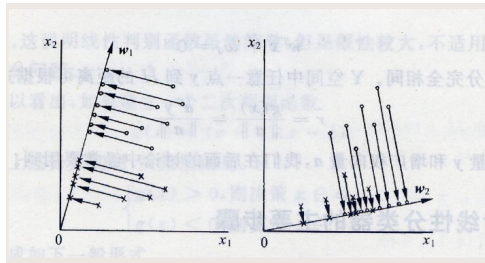
Only the direction of projection changes.

Poor direction

Different classes land on overlapping positions.

Useful direction

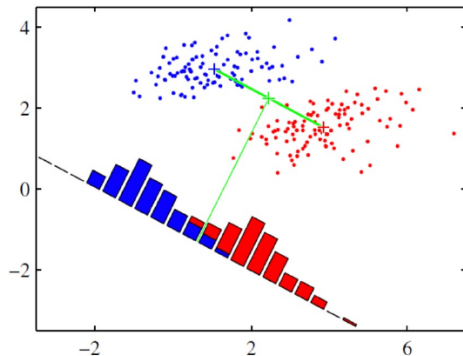
Class centers separate and within class spread stays small.



The projected histograms overlap strongly.

The axis preserved a direction with large within class variation.

Lesson. Separation in the original plane does not survive every one dimensional view.



A useful projection separates the classes

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Visible result

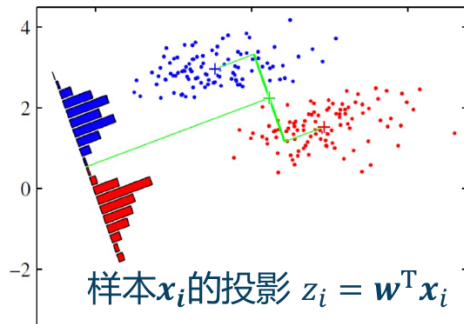
The projected histograms occupy different parts of the axis.

Why it works

The direction keeps center separation while reducing within class spread.

No guarantee of zero error

Real classes may still overlap after the best linear projection.



$\langle \cdot, \cdot \rangle$ is an inner product

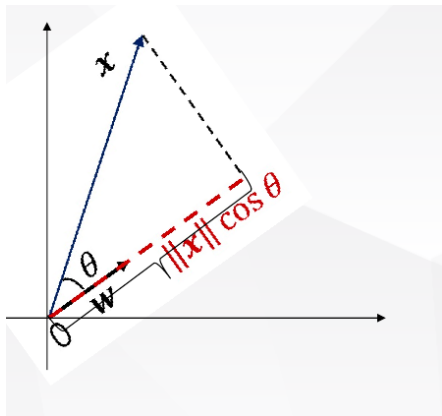
Unit direction

If $\|w\| = 1$, then $z = w^\top x = \|x\| \cos \theta$.

Only direction matters

Replacing w by cw rescales every projected value by c .

Geometry. The perpendicular foot on the axis determines the scalar coordinate.



$$m_k = \frac{1}{N_k} \sum_{i: y_i = k} x_i$$

$$\tilde{m}_k = \frac{1}{N_k} \sum_{i: y_i=k} w^\top x_i = w^\top m_k$$

Projected center difference is $w^\top (m_1 - m_2)$.

Projected within class spread

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Class scatter

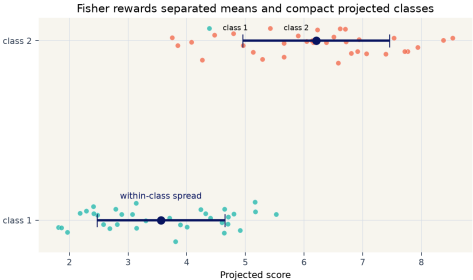
$$S_k = \sum_{i: y_i=k} (x_i - m_k)(x_i - m_k)^T$$

Projected scatter

$$\tilde{S}_k^2 = \sum_{i: y_i=k} (w^T x_i - w^T m_k)^2 = w^T S_k w$$

Total within scatter

$w^T S_W w$ where $S_W = S_1 + S_2$.



class separation

Scalar separation

$$(\tilde{m}_1 - \tilde{m}_2)^2 = [w^\top (m_1 - m_2)]^2$$

Matrix form

$$S_B = (m_1 - m_2)(m_1 - m_2)^\top$$

$$[w^\top (m_1 - m_2)]^2 = w^\top S_B w$$

er criterion is a ratio

Large numerator

Projected class means should be far apart.

Small denominator

Projected samples within each class should stay compact.

$$J(w) = \frac{w^\top S_B w}{w^\top S_W w} = \frac{[w^\top (m_1 - m_2)]^2}{w^\top S_W w}$$

The Fisher ratio ignores scale

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Rescale the direction

Let $w' = cw$ for any nonzero c .

$$J(cw) = \frac{c^2 w^\top S_B w}{c^2 w^\top S_W w} = J(w)$$



Practical normalization

Use $w/\|w\|$ for display, or fix $w^\top S_W w = 1$ in a derivation.

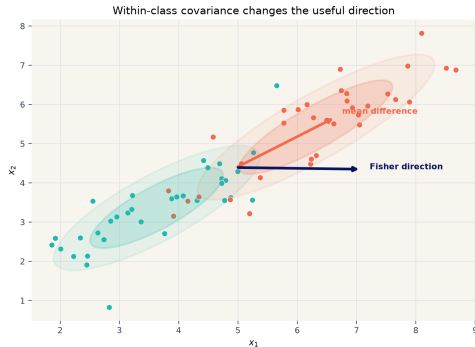
$$W = m_1 - m_2$$

Problem

It ignores directions where each class has large variance.

Elongated clouds

The center line may align with a noisy long axis and create overlapping projections.



$$w \propto S_W^{-1}(m_1 - m_2)$$

The inverse scatter reduces its contribution.

Low variance direction

A modest mean difference can matter when measurements are consistent.

Scatter matrices come from centered samples

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

- 1 Compute m_1 and m_2 from their own classes.
- 2 Center every sample using its class mean.
- 3 Accumulate outer products inside each class.
- 4 Add the two matrices to obtain S_W .



Common mistake

Centering all samples by the global mean mixes between class separation into the within class scatter.

Solve a linear system instead of an inverse

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Mathematical expression

$$w = S_W^{-1}(m_1 - m_2)$$

Numerical computation

Solve $S_W w = m_1 - m_2$.

```
Sw = np.array([[4., 1.], [1., 2.]])
mean1 = np.array([1., 1.])
mean2 = np.array([3., 2.])
w = np.linalg.solve(Sw, mean1 - mean2)
```

Reason

Explicit matrix inversion is slower and usually less numerically stable.

A two dimensional Fisher example

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Class means

$$m_1 = (2, 0)^T \text{ and } m_2 = (2, 2)^T$$

Within class scatters

$$S_1 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix} \text{ and } S_2 = \begin{bmatrix} 1 & -1/2 \\ -1/2 & 1 \end{bmatrix}$$

$$S_W = 2I, \quad w = S_W^{-1}(m_1 - m_2) = (0, -1)^T$$

$$t = (0 - 2)/2 = -1.$$

Fisher selects w to separate one dimensional class distributions.

Choose a threshold t and compare $z = w^\top x$ with t .

The midpoint $(w^\top m_1 + w^\top m_2)/2$ is a common choice.

The Fisher criterion alone does not encode priors or unequal error costs.

Class priors move the projected threshold

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Equal projected variance

If $z \mid k \sim \mathcal{N}(\mu_k, \sigma^2)$, Bayes gives

$$t = \frac{\mu_1 + \mu_2}{2} - \frac{\sigma^2}{\mu_1 - \mu_2} \log \frac{\pi_1}{\pi_2}$$



Interpretation

A more common class receives a larger region along the projection axis.

Fisher and equal covariance Gaussian discrimination

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Shared covariance model

$$x \mid y = k \sim \mathcal{N}(\mu_k, \Sigma)$$

$$w_{\text{Gaussian}} = \Sigma^{-1}(\mu_1 - \mu_2)$$



Direction

Fisher uses $S_W^{-1}(m_1 - m_2)$, which has the same form.

Threshold

Priors and losses still determine the intercept of the Gaussian decision rule.

Fall 2026 | 100 / 108

s Fisher uses several axes

Within class scatter

Between class scatter

At most $C - 1$ discriminative axes have nonzero between class information.

Fisher direction in NumPy

```
X1 = np.array([[1., 1.], [2., 0.], [2., 1.], [3., 0.]])
X2 = np.array([[1., 3.], [2., 2.], [2., 3.], [3., 2.]])
m1, m2 = X1.mean(axis=0), X2.mean(axis=0)
```

```
C1, C2 = X1 - m1, X2 - m2
Sw = C1.T @ C1 + C2.T @ C2
w = np.linalg.solve(Sw + 1e-6*np.eye(2), m1 - m2)
w = w / np.linalg.norm(w)
```

```
z1, z2 = X1 @ w, X2 @ w
threshold = 0.5 * (z1.mean() + z2.mean())
print(m1, m2, Sw, w, threshold, sep='\n')
```

Expected direction

The normalized vector points almost straight down.

Model checks 数字工程研究中心

- 1 Compare training and validation separation on the projected axis.
- 2 Inspect projected distributions rather than only their means.
- 3 Report the threshold, priors, and loss assumptions separately.
- 4 Stress test the direction after small data and scaling changes.

A high criterion is not enough

A few outliers or a shifted deployment population can still produce poor decisions.

Fisher checkpoint与数字工程研究中心

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Given

$$m_1 = (1, 1)^\top, m_2 = (3, 2)^\top, S_W = \text{diag}(4, 1)$$

- 1 Compute $m_1 - m_2$.
- 2 Solve for the Fisher direction.
- 3 Explain why the second feature receives more relative weight.



Answer

$m_1 - m_2 = (-2, -1)^\top$, so $w \propto (-0.5, -1)^\top$. The first feature has four times the within class scatter.

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

Design	What is learned	Best first question
Linear	one score $w^\top x + w_0$	Can one straight boundary work?
Feature map	a linear score in $\phi(x)$	Which transformation exposes the pattern?
Piecewise linear	several local linear scores	Does the boundary need several faces?
Fisher	a projection with low within class spread	Which direction separates compactly?

Begin with the simplest geometry that can express the decision.

Size separates most samples, but two growing regions create a bent boundary in color and firmness.

Choose the threshold from the cost of sending a wrong variety to packaging.

From geometry to learning algorithms

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

This lecture

We designed decision functions and understood their geometry.

Next lecture

The perceptron will learn a separating direction from mistakes; decision trees will partition the space by questions.

Same data, different inductive bias

A learning rule is not only an optimizer. It also determines which boundaries are easy to discover.

Exit ticket 先进计算与数字工程研究中心

ADVANCED COMPUTING AND DIGITAL TECHNOLOGY RESEARCH CENTER

- 1 State the difference between a generative and a discriminative classifier in one sentence.
- 2 Explain the sign and magnitude of a linear discriminant score.
- 3 Give one reason to use a feature map and one reason to use linear pieces.
- 4 Explain the numerator and denominator of the Fisher criterion.
- 5 For XOR, choose a model family and justify the choice.



A strong answer

Names the geometry, the information used, and one failure mode rather than listing formulas.